

in Barcelona than in Andalusia. However, the correlation between hate speech and polarization for the detrended data is not statistically significant (see Table 6).

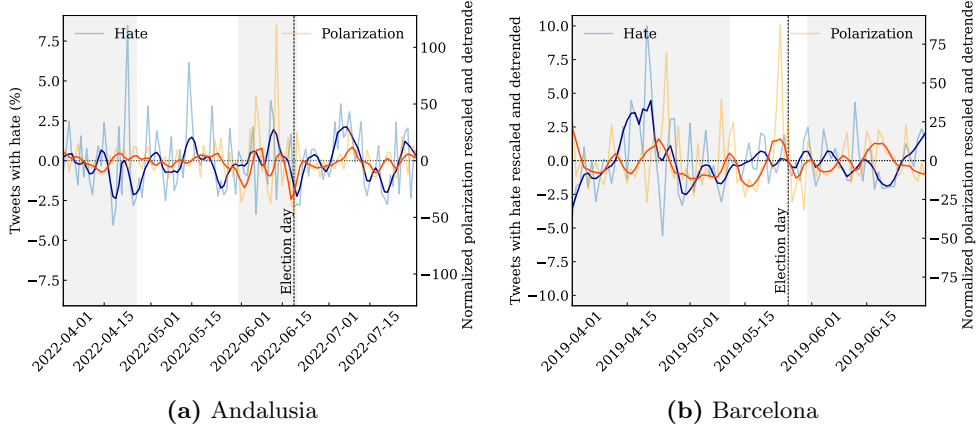


Fig. 7 Comparison between the detrended and rescaled percentage of tweets with hate speech and polarization of the interaction network for each day. The grey shadow represents the period between change points.

Region	Without detrending	Detrended data
Andalusia	-0.3113 (0.0005)	-0.1456 (0.1)
Barcelona	-0.3306 (0.001)	-0.1817 (0.08)

Table 6 PCC of polarization and hate for total period data with and without detrending.

3.4 Stability of communities

Now we calculate the probability that a user changes community from one day to the next. We can identify fluxes between communities and its stability in Fig. 8. An extended version of this figure with all the communities detected by the algorithm can be found in the Supplementary Material (Figs. S1.1 and S1.2). For Andalusia we see that the interchange of users happens especially inside left-wing parties: PorA and PSOE-A. There is a probability that AA users move to PorA, but not the opposite. Also, there is a probability that PP and Cs communities users move to Vox. However, there is little probability that a Vox community user move to another community.

For Barcelona, we can see a high probability of moving from community to another for users from BComú and PSC. There is also some interchange between (in order) CUP to Junts, ERC to Junts, CUP to ERC, Junts to ERC and Junts to CUP. Here, Cs community is the most stable by difference, with almost no probability to exchange users.

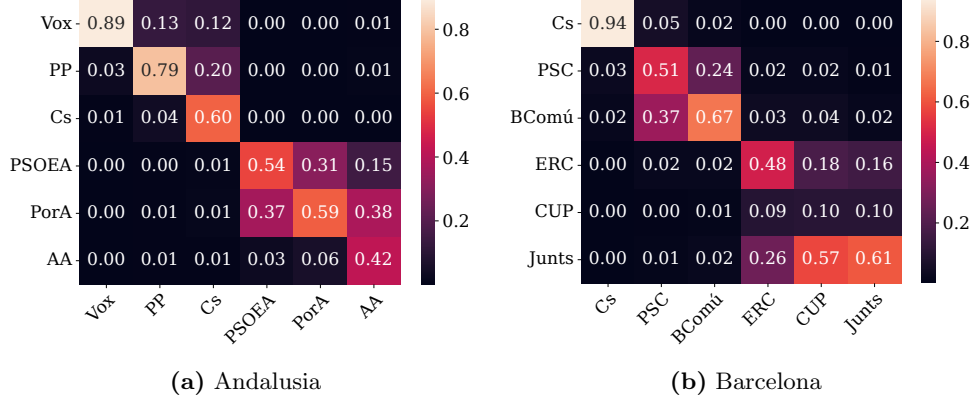


Fig. 8 Estimation of probability of a user belonging to community X in a day to become a member of the community Y the day after.

4 Discussion

This research advances our understanding of political polarization during electoral campaigns in multi-party systems, providing insights into how these dynamics unfold, particularly in the presence of new political actors. Our findings reveal significant differences in polarization patterns between elections with and without the presence of a far-right party with a realistic chance of winning.

In the case of Andalusia, where the far-right party Vox was a significant contender, we observed a clear growing trend of structural polarization throughout the electoral campaign, peaking on election day and followed by a rapid drop. The observed pattern aligns with the trend of user engagement in the political discussion. This result is in line with other studies on polarization in dichotomous political campaigns, that used different methodologies to measure polarization (Olivares et al., 2019).

Moreover, our analysis of the community structure of the retweet network highlights a pronounced two-block arrangement, with users clustering around right-wing versus left-wing ideologies. This indicates that, despite the presence of multiple parties, the election dynamics in Andalusia exhibited characteristics of block polarization (Hansen and Kosiara-Pedersen, 2017), effectively rendering the campaign dichotomous. The existence of these ideological blocks is further corroborated by the analysis of party change probabilities for users from day to day, which indicates a higher likelihood of shifts occurring within the same ideological group, reinforcing the two-block structure. This tendency is higher for users in the left-wing block.

The picture is different in the case of the Barcelona city council elections. Unlike in Andalusia, where the left-right polarization axis dominates, the political landscape in Barcelona is significantly perturbed by the pro-/anti-independence axis. This dual axis of polarization reshapes the retweet network structure.

In the case of Barcelona, Spanish right-wing anti-independence parties form a unique community, which is connected to the rest of the network primarily through the left-wing anti-independence party, PSC. Interestingly, the radical left pro-independence party CUP is positioned close to the centre-right pro-independence party

Junts, indicating a convergence around the independence issue. The left-wing populist and moderately anti-independence party BComú is situated near both CUP and PSC, highlighting its intermediary role between the pro- and anti-independence camps.

Further analysis of party change probabilities reveals a three-block structure: the Spanish right-wing, left-wing anti-independence, and pro-independence groups. Political polarization is generally considered higher when positions on different issues align closely with party choice (DiMaggio et al., 1996). In this case, however, the presence of multiple intersecting issues dilutes the alignment, leading to lower overall polarization. Structural polarization metrics capture this complexity, with average polarization in Andalusia being notably higher compared to the lower levels observed in Barcelona.

The temporal evolution of polarization in Barcelona reveals that the dynamics are heavily influenced by specific events external to the campaign, such as the Spanish general election debate. On these key days, polarization spikes significantly, disrupting any clear overall trend of increasing polarization throughout the campaign. Consequently, in Barcelona, the largest networks do not correspond to the most polarized moments. This contrasts with the more consistent growth of polarization observed in Andalusia, where the campaign’s progression steadily intensified polarization up until election day.

These findings suggest that the presence of a far-right party with a realistic chance of winning significantly polarizes the electoral discussion, driving it into two distinct ideological blocks. In contrast, a dichotomous issue like independence does not lead to the same level of heightened polarization, as other important political stances, which are transversal, remain relevant and prevent a complete alignment of the electorate into just two opposing groups.

Turning to the analysis of hate speech detection, we find that the average percentage of tweets containing hate speech is 14.9 ± 0.2 in Andalusia and 13.1 ± 0.3 in Barcelona. Both in Andalusia and Barcelona, polarization and hate speech are anti-correlated before the election date, while this correlation is not statistically significant afterward. Notably, in both cases, the anti-correlation disappears after detrending, indicating that the fluctuations in polarization and hate speech are not directly correlated.

In Andalusia, the anti-correlation between polarization and hate speech before the election can be explained by the increasing participation of users from the block opposing Vox. As more users from this opposing block engage in the discussion, overall polarization rises, but the prominence of hate speech tweets decreases, since these users tend not to employ hate speech in their discourse. Additionally, Capdevila et al. (2022) have suggested that far-right parties like Vox may moderate their speech during electoral campaigns due to the pressure of Spanish electoral law.

A key limitation of this study is the difference in scope between regional and city council elections. While regional elections often address broader issues, municipal elections focus on local concerns. Nevertheless, city council elections can still generate significant attention, especially in cities with complex political landscape.

In summary, our findings underscore the significant role that populist far-right actors play in shaping polarization patterns. While far-right parties tend to polarize